

## Session 5.1

# Text as Economic Data: From Documents to Counts

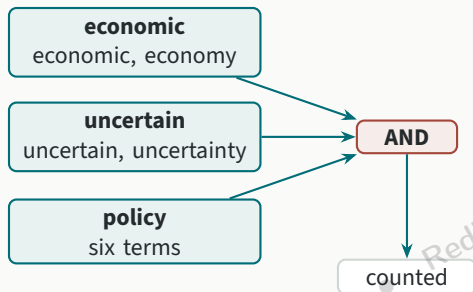
Clean, tokenize, count: how a document becomes a vector a regression can use

Based on Chapter 13 of Miao & Feng, *AI for Economists*



Zhigang Feng

# Reading Policy Uncertainty Out of Newspapers

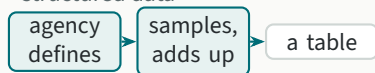


an article counts only if it hits all three groups

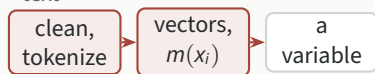
- ▶ Baker, Bloom & Davis (2016) read policy uncertainty from ten US newspapers, month by month since 1985.
- ▶ With  $c_{pt}$  hits among  $N_{pt}$  articles of paper  $p$  in month  $t$ :  $U_t = \frac{1}{10} \sum_p \frac{c_{pt}/N_{pt}}{\sigma_p}$ , and the index is  $100 U_t / \bar{U}$ .
- ▶ The six policy words were chosen against 12,009 hand-coded articles. The same recipe gives a China index from 1949 (Davis, Liu & Sheng 2019).
- ▶ **No agency publishes this number: it was built from words.**

# With Text, the Measurement Step Is Yours

structured data



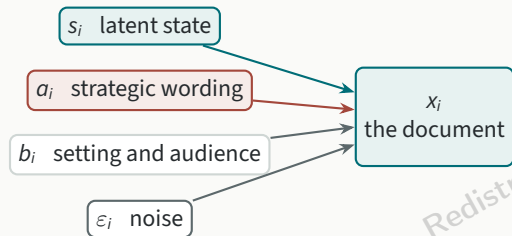
text



the same regression at the end; who did the measuring differs

- ▶ Prices and output arrive measured; FOMC statements, 10-Ks, earnings calls and job ads arrive as language.
- ▶ For a price index the agency defines, samples and adds up. For text nobody has done it.
- ▶ A text variable can be a measure, an outcome or a treatment; this session builds the first.
- ▶ **With text, the researcher does the measuring.**

# What a Text Measure Measures



two firms, the same risk  $s_i$ : one discloses,  
one defects, so  $x_i$  differs through  $a_i$

- ▶ The document is not the state you want:  $x_i = g(s_i, a_i, b_i) + \varepsilon_i$ .
- ▶ The measure  $m(x_i)$  enters an ordinary regression:  $y_i = \beta m(x_i) + \gamma^\top \mathbf{c}_i + u_i$ .
- ▶ If wording  $a_i$  and the error  $u_i$  share a driver,  $\hat{\beta}$  is biased; a bigger corpus does not repair it.
- ▶ **The error moves with the state, so its sign is unknown: ask why  $m(x_i)$  is exogenous.**

# The Pipeline: Four Steps From a String to a Vector



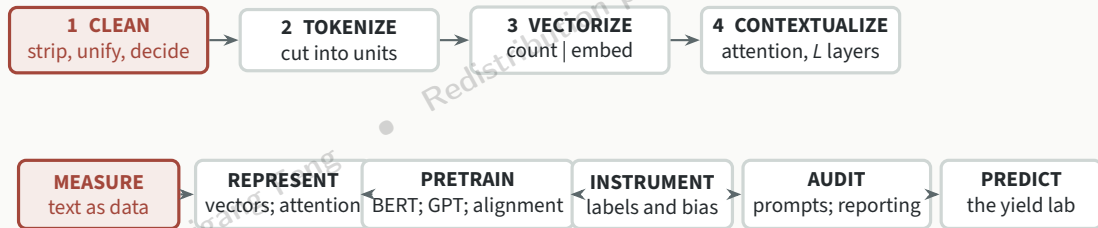
|    | after step 1: clean                                           | after step 2: tokenize                                                           | after step 3: count             |
|----|---------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------|
| EN | The central bank raised interest rates amid rising inflation. | The · central · bank · raised · interest · rates · amid · rising · inflation · . | one row of $V$ counts, mostly 0 |
| ZH | 央行上调利率以遏制通胀。                                                  | 央行 · 上调 · 利率 · 以 · 遏制 · 通胀 · 。                                                   | one row of $V$ counts, mostly 0 |

- ▶ Cleaning and tokenizing fix which strings count as one feature, the coordinates every later model works in. Vectorizing turns tokens into numbers: this deck counts; Deck 5.2 embeds and adds context.
- ▶ **Different choices at any step give different regressors from the same documents.**

## Session 5.1

# Step 1 • Clean

What to remove, what to unify, and what must stay



## Cleaning, Worked on One Sentence

English, scraped from a web page

`<p>The central bank&nbsp;raised ...</p>`

↓ strip mark-up and entities

The central bank raised ...

Chinese, traditional characters

`<p>央行上調利率以遏制通脹。</p>`

↓ 繁 → 简: 調 → 调, 脹 → 胀

央行上调利率以遏制通胀。

C P I 同比上漲 2 . 3 %

↓ 全角 → 半角, 漲 → 涨

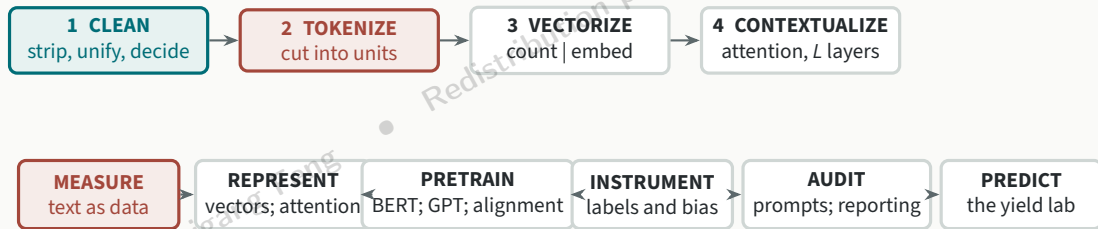
CPI 同比上涨 2.3%

- ▶ Strip what is not text: mark-up, page headers and footers, table debris, broken encodings.
- ▶ Unify what should be one string. Otherwise 通脹 and 通胀 are two words, and C P I and CPI never match.
- ▶ Decide by task: lower-casing merges the Fed with the verb *fed*; in finance %, ? and quotation marks carry information.
- ▶ **Write the rules as a script, and archive it with the data.**

## Session 5.1

# Step 2 • Tokenize

Cut the string into the units you will count



# Tokens and the Vocabulary

English: spaces do most of the cutting

GDP rose by 2 %

Chinese: the boundaries must be inferred

央行 决定 下调 存款准备金率

央行决定下调存款准备金率: the central bank decides to cut the reserve requirement ratio

- ▶ A **corpus** is a collection of texts: a year of FOMC minutes, every annual report in an industry, 一个季度的财经新闻 (a quarter of financial news).
- ▶ A **token** is the smallest unit you count, and **tokenization** cuts a character string into tokens. The **vocabulary** is the set of distinct tokens.
- ▶ **Change the cut and you change every count that follows.**

EN The central bank raised interest rates amid rising inflation .

ZH 央行 上调 利率 以 遏制 通胀 。

the running pair, cut into words

# Chinese Segmentation Is a Labelling Problem

研究生生命科学

研究 生命科学

study | life science

研究生 命科学

graduate student | ?

行业政策支持

行业政策 支持

head word: support

行业 政策支持

head word: policy support

one label per character: B begins a word, E ends it

央 行 上 调 利 率

B E B E B E

- ▶ Chinese characters run together, so segmentation is **sequence labelling**: for each character, decide whether a word ends after it.
- ▶ 研究生命科学 reads as *study life science*, or as nonsense if the cut falls after 生.
- ▶ 行业政策支持 splits two sensible ways with different head words: support for an industry policy, or policy support for an industry.
- ▶ **A wrong boundary changes what the sentence is about.**

# Domain Compounds Need a User Dictionary

货币政策委员会, the monetary policy committee  
general dictionary

货币 政策委员会 head word lost

with a user dictionary

货币政策委员会 kept whole

entries to add first:

社会融资规模 宏观审慎监管框架

- ▶ Segmenters such as Jieba, THULAC and LTP combine a dictionary with a statistical or neural labeller.
- ▶ Domain compounds are missing from general dictionaries, and a wrong cut changes the head word.
- ▶ The fix is a **user dictionary** listing terms such as 社会融资规模 (aggregate financing), so they stay whole.
- ▶ **Add the domain terms before you count, and publish the list.**

## Stop Words: Drop the Glue, Keep the Signal

usually dropped

the is of 的 了 是

keep: they carry the signal

not up down more less

因此 therefore 然而 however

- ▶ **Stop words** are frequent function words with little topical content: the, is, of; 的, 了, 是.
- ▶ General lists also delete up, down, more and less, the direction of a macro signal; deleting *not* flips a sentence. 因此 and 然而 carry the logic.
- ▶ Domain matters too: a general sentiment list marks *liability* negative, but in a set of accounts it is a neutral line item (Loughran & McDonald 2011).
- ▶ **A general stop list can delete the signal itself.**

# Stems, Lemmas, and a Language Without Inflection

stemming: suffixes cut by rule

investing, invests, invested → invest

studies → studi

lemmatization: a dictionary form

better → good

were → be

Chinese: the verb never changes

已 上调      将 上调

- ▶ **Stemming** chops suffixes by rule, so investing, invests and invested all become *invest*; it also turns studies into *studi*, a non-word.
- ▶ **Lemmatization** returns the dictionary form using the part of speech: better → good, were → be.
- ▶ 上调 (raise) never changes form: tense sits in separate words, 已上调 (has raised) and 将上调 (will raise).
- ▶ **English must choose stem or lemma; Chinese has no need for either.**

# n-Grams: One Boundary Can Flip the Sign

scheme A: one token at a time

not sufficient 并未 放缓

“sufficient” and 放缓 (slowing) count on their own, so the sentence’s sign can invert

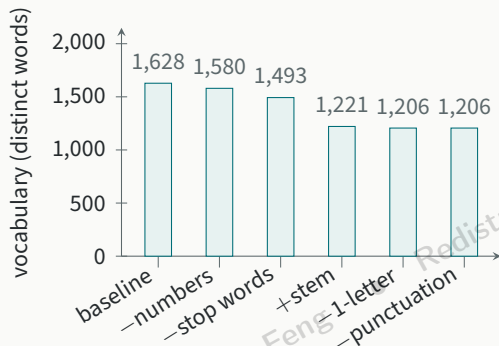
scheme B: adjacent pairs

not\_sufficient 并未\_放缓

each negation survives as one feature

- ▶ An **n-gram** treats  $n$  adjacent tokens as one feature: unigrams are single tokens, bigrams adjacent pairs.
- ▶ Pairs keep compounds and negations whole: central\_bank; 货币政策 (monetary policy); 并未\_放缓 (has not slowed).
- ▶ Under unigrams a hawkish “not sufficient” can read as reassuring; Hassan et al. (2019) count word pairs to measure political risk.
- ▶ **Pairs keep the relation; the price is a much larger vocabulary.**

## Six Pipelines, Five Vocabularies



The last two bars coincide: this tokenizer never emits punctuation, so there are 5 vocabularies, not six.

- ▶ The lab's corpus: 245 FOMC statements, 1994–2026; the 149 from 1994–2014 are used for fitting.
- ▶ Six defensible pipelines on those, each adding one step: 1,628 → 1,206 words (−26%).
- ▶ “...to 5-1/4 percent” loses the rate itself; “has moderated *but* remains elevated” loses its turn: infl · moderat · remain · elevat.
- ▶ **Report your pipeline; show the result survives a change** (Denny & Spirling 2018).

# Why Subwords

a word split into reusable pieces

macroeconomics → macro economics

not among the lab's 1,628 words, yet  
spelled by its byte-pair pieces:

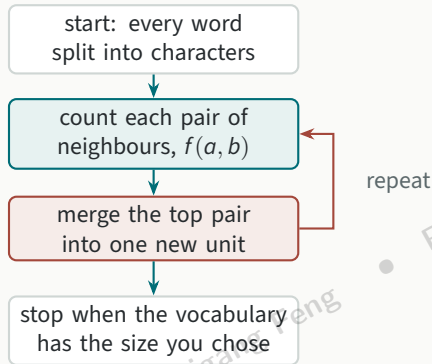
di·s·infl·a·tion·ar·y      disinflationary

a policy term coined after the dictionary:

专精特新 → 专 精 特 新

- ▶ Word-level tokenizers fail twice: a word never seen in training gets no representation, and the vocabulary grows with every new form.
- ▶ **Subword tokenization** keeps reusable fragments instead, so a new word is spelled from known pieces.
- ▶ Firm names, tickers and new policy terms such as 专精特新 or 稳健中性 are exactly the words a dictionary misses.
- ▶ **With byte-level subwords, every string has a representation.**

# Byte-Pair Encoding: Learn the Pieces From Counts



- ▶ Goal: a vocabulary of a chosen size; frequent strings are one token, rare ones are spelled from pieces (Sennrich, Haddow & Birch 2016).
- ▶ Start from the characters. Count each adjacent pair,  $f(a, b) = \sum_w n_w c_{ab}(w)$ : word  $w$ 's count times the pair's occurrences in  $w$ .
- ▶ Merge  $(a^*, b^*) = \arg \max f(a, b)$  into a new unit, rewrite every word, count again.
- ▶ **The output is an ordered list of merges. To cut any word, replay them in order.**

## BPE Worked: Four Merges on Four Words

a four-word corpus: 通胀 inflation · 通缩 deflation · 通胀率 inflation rate · 利率 interest rate

| word | count | start | merge 1      | merge 2      | merge 3       | merge 4      |
|------|-------|-------|--------------|--------------|---------------|--------------|
|      |       |       | 通+胀, $f = 8$ | 利+率, $f = 4$ | 通胀+率, $f = 3$ | 通+缩, $f = 2$ |
| 通胀   | 5     | 通 胀   | 通胀           | 通胀           | 通胀            | 通胀           |
| 通缩   | 2     | 通 缩   | 通 缩          | 通 缩          | 通 缩           | 通缩           |
| 通胀率  | 3     | 通 胀 率 | 通胀 率         | 通胀 率         | 通胀率           | 通胀率          |
| 利率   | 4     | 利 率   | 利 率          | 利率           | 利率            | 利率           |

- Merge 1, 通+胀: 5 times in 通胀 plus 3 in 通胀率,  $f = 8$ . Each column counts again on the rewritten words.
- A new word reuses the merges: 通缩率 (deflation rate)  $\rightarrow$  通缩 · 率; after two merges 通缩 is still 通 · 缩.
- On the FOMC statements, 200 merges give un·emplo·y·ment. **Frequency, not meaning, decides the cuts.**

# Three Ways to Learn the Pieces

|                                                                           |      |      |         |          |           |           |     |       |       |        |
|---------------------------------------------------------------------------|------|------|---------|----------|-----------|-----------|-----|-------|-------|--------|
| one phrase, “disinflationary pressures eased”, as three tokenizers cut it |      |      |         |          |           |           |     |       |       | pieces |
| byte-pair (GPT-4’s cl100k)                                                | dis  | in   | flation | ary      | ▮         | pressures | ▮   | eased |       | 6      |
| WordPiece (multilingual BERT)                                             | dis  | ##in | ##f     | ##lation | ##ary     | pressure  | ##s | ea    | ##sed | 9      |
| SentencePiece (XLM-R)                                                     | _dis | inf  | lation  | ary      | _pressure | s         | _ea | sed   |       | 8      |

| scheme                                         | how a piece enters the vocabulary          | word boundary          | used by                   |
|------------------------------------------------|--------------------------------------------|------------------------|---------------------------|
| byte-pair (Sennrich et al. 2016)               | merge the most frequent pair, on bytes     | space inside the piece | GPT-2 to 4, LLaMA 3       |
| WordPiece (Schuster & Nakajima 2012)           | merge the pair that most raises likelihood | ## on later pieces     | BERT (Devlin et al. 2019) |
| SentencePiece unigram (Kudo & Richardson 2018) | prune a large candidate set by likelihood  | _ on a first piece     | T5, XLM-R                 |

► Each learned its pieces on its own corpus: one phrase, three cuts.

# One Sentence, Five Tokenizers

the Chinese sentence, final 。 included, as each tokenizer cuts it

tokens: ZH / EN

|                    |                                                             |         |
|--------------------|-------------------------------------------------------------|---------|
| cl100k (GPT-4)     | E5 A4 AE 行 上 调 利 率 以 E9 81 8F 制 通 E8 83 80 。                | 15 / 10 |
| o200k (GPT-4o)     | 央 行 上 调 利 率 以 E9 81 8F 制 通 E8 83 80 。                       | 14 / 10 |
| BERT, English      | [UNK] 行 上 [UNK] [UNK] [UNK] [UNK] [UNK] [UNK] [UNK] [UNK] 。 | 12 / 10 |
| BERT, Chinese      | 央 行 上 调 利 率 以 遏 制 通 胀 。                                     | 12 / 20 |
| BERT, multilingual | 央 行 上 调 利 率 以 遏 制 通 胀 。                                     | 12 / 11 |

- ▶ Byte-level BPE splits a character it saw rarely into raw UTF-8 bytes (red, in hex): 央 is three bytes, cut in two by cl100k. The newer o200k keeps 央 whole but still splits 遏 and 胀.
- ▶ English BERT knows few Chinese characters: 9 of its 12 tokens are [UNK]. Chinese BERT cuts every character, and cuts the English sentence into 20 pieces.
- ▶ **The same sentence is 10 tokens to one model and 20 to another.**

# The Tokenizer Belongs to the Model

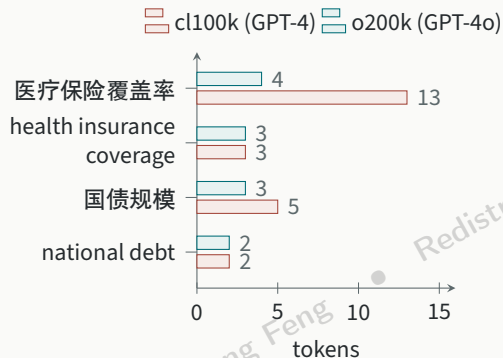
社会融资规模, cut four ways

|               |    |    |       |       |       |       |   |   |
|---------------|----|----|-------|-------|-------|-------|---|---|
| o200k         | 社会 | 融资 | 规模    |       |       |       |   |   |
| cl100k        | 社  | 会  | E8    | 9E    | 8D    | 资     | 规 | 模 |
| BERT, Chinese | 社  | 会  | 融     | 资     | 规     | 模     |   |   |
| BERT, English | 社  | 会  | [UNK] | [UNK] | [UNK] | [UNK] |   |   |

aggregate financing to the real economy; in  
cl100k, 融 is three raw bytes

- ▶ One term, four cuts: 3 pieces, 8, 6 characters, or [UNK]s. “GDP rose by 2%”, 5 tokens cut by words, is 6 or 7 GPT tokens.
- ▶ A model’s first layer is an **embedding table**: one learned vector per token ID. Feed it another tokenizer’s IDs and every row is the wrong vector.
- ▶ Mixed Chinese–English notes and filings that mix 简 and 繁 call for byte-level BPE.
- ▶ **Report the tokenizer, and its version, as you report the sample period.**

# Translate, or Read in the Original?

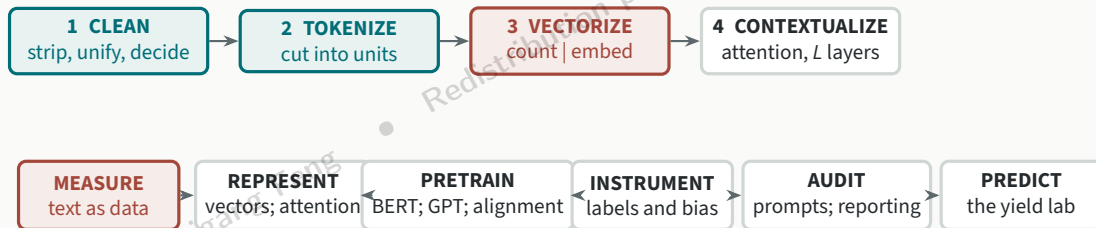


- ▶ One idea need not map one to one: 稳健 means both stable and sound, while “prudent” keeps only caution; the colour of 适度宽松 is hard to render.
- ▶ Translating drifts the colour; a multilingual model keeps the wording, with quality that varies by language.
- ▶ Chinese costs more tokens (left), so it fills a model’s window sooner.
- ▶ **If a conclusion rests on a hard-to-translate phrase, report both routes.**

## Session 5.1

# Step 3 • Count

Turn tokens into vectors: one axis per word



## From Tokens to Numbers: IDs and One-Hot Vectors

vocabulary, numbered

|                            | fed | cut | hike | support | curb | growth | inflation | respond |
|----------------------------|-----|-----|------|---------|------|--------|-----------|---------|
| ID                         | 1   | 2   | 3    | 4       | 5    | 6      | 7         | 8       |
| $\mathbf{o}_{\text{fed}}$  | 1   | 0   | 0    | 0       | 0    | 0      | 0         | 0       |
| $\mathbf{o}_{\text{hike}}$ | 0   | 0   | 1    | 0       | 0    | 0      | 0         | 0       |

$\mathbf{o}_{\text{fed}}^T \mathbf{o}_{\text{hike}} = 0$ : no two different words share a coordinate

- ▶ Number the vocabulary: each distinct token gets an ID  $v = 1, \dots, V$ .
- ▶ Token  $v$  becomes the **one-hot vector**  $\mathbf{o}_v \in \{0, 1\}^V$ : a 1 in place  $v$ , 0 everywhere else.
- ▶ For two different words,  $\mathbf{o}_v^T \mathbf{o}_{v'} = 0$ : every pair of words is **orthogonal** by construction, whatever the words mean. *hike* and *raise* are as far apart as *hike* and *cut*.
- ▶ **A document is the sum of its one-hot vectors: a row of counts.**

# Bag of Words and the Document-Term Matrix

|                | fed | cut | hike | support | curb | growth | inflation | respond |
|----------------|-----|-----|------|---------|------|--------|-----------|---------|
| $\mathbf{z}_1$ | 1   | 1   | 0    | 1       | 0    | 1      | 0         | 0       |
| $\mathbf{z}_2$ | 1   | 0   | 1    | 0       | 1    | 0      | 1         | 0       |
| $\mathbf{z}_3$ | 1   | 0   | 1    | 0       | 0    | 0      | 2         | 1       |

three documents,  $V = 8$ :  $\mathbf{Z} \in \mathbb{R}^{3 \times 8}$ ,  
frequency counts

- ▶ Three short documents, already cleaned and tokenized:  $d_1 = \text{fed cut support growth}$ ;  $d_2 = \text{fed hike curb inflation}$ ;  $d_3 = \text{fed hike respond inflation inflation}$ .
- ▶ Count each word and ignore order: the **document-term matrix**  $\mathbf{Z} \in \mathbb{R}^{D \times V}$ , one row per document.
- ▶ Binary counts record presence (inflation twice in  $d_3$  gives 1); frequency counts give 2.
- ▶ **Each row is its document's vector  $\mathbf{z}_i$ .**

## TF-IDF, Worked Through

| term      | $df_v$ | $idf_v = \log(D/df_v)$ | $tf_{v,3}$ | TF-IDF in document 3 |
|-----------|--------|------------------------|------------|----------------------|
| fed       | 3      | $\log(3/3) = 0$        | 1          | 0                    |
| hike      | 2      | $\log(3/2) = 0.405$    | 1          | 0.405                |
| inflation | 2      | $\log(3/2) = 0.405$    | 2          | 0.811                |
| respond   | 1      | $\log(3/1) = 1.099$    | 1          | 1.099                |

- ▶  $TF\text{-}IDF_{v,i} = tf_{v,i} \times \log(D/df_v)$ , with  $tf_{v,i}$  the count of word  $v$  in document  $i$  and  $df_v$  the number of documents containing it: frequent here, rare across the corpus.  $d_3$  becomes  $(0, 0, 0.405, 0, 0, 0, 0.811, 1.099)$ .
- ▶ A graded stop list: boilerplate such as “financial statements” has  $df \approx D$  and a weight near 0, with no list written in advance.
- ▶ **The trap: if every statement contains “not”, its weight goes to zero too.**

# The Bag Forgets Word Order

GDP growth exceeded inflation

inflation exceeded GDP growth

both become the same count vector:

| GDP | growth | exceeded | inflation |
|-----|--------|----------|-----------|
| 1   | 1      | 1        | 1         |

the rate cut supports growth

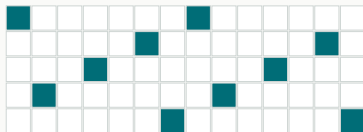
growth supports the rate cut

the same bag

- ▶ “GDP growth exceeded inflation” and “inflation exceeded GDP growth” use the same words, so they get the same vector.
- ▶ “The rate cut supports growth” and “growth supports the rate cut” state opposite causal directions, in identical bags.
- ▶ Harmless when you count topic words, as the uncertainty index does; fatal when direction, cause or negation is the object.
- ▶ **Word order is gone. Deck 5.2 puts it back with position vectors.**

## Sparse: Most of the Matrix Is Zero

a document-term matrix, schematic



filled cells are non-zero; in the lab's 245  
× 1,163 TF-IDF matrix, 85.5% of the  
cells are 0

- ▶ In the three-document example, each row has 4 non-zero entries out of 8. With tens of thousands of words, often fewer than one cell in a hundred is non-zero.
- ▶ With fewer documents than words ( $D \ll V$ ), OLS on all  $V$  columns is not identified:  $\mathbf{Z}^\top \mathbf{Z}$  cannot be inverted.
- ▶ Reduce the dimension, with topic models, or penalise, with the Lasso.
- ▶ **That is why text regressions borrow the tools of machine learning.**

## Comparing Documents: Cosine on Counts

|        | <i>fed</i> | <i>cut</i> | <i>hike</i> | <i>support</i> | <i>curb</i> | <i>growth</i> | <i>inflation</i> | <i>respond</i> | <i>raise</i> |
|--------|------------|------------|-------------|----------------|-------------|---------------|------------------|----------------|--------------|
| $d_1$  | 1          | 1          | 0           | 1              | 0           | 1             | 0                | 0              | 0            |
| $d_2$  | 1          | 0          | 1           | 0              | 1           | 0             | 1                | 0              | 0            |
| $d_3$  | 1          | 0          | 1           | 0              | 0           | 0             | 2                | 1              | 0            |
| $d'_2$ | 1          | 0          | 0           | 0              | 1           | 0             | 1                | 0              | 1            |

$$\cos(d_1, d_2) = 1/(2 \cdot 2) = 0.25$$

$$\cos(d_2, d_3) = (1 + 1 + 2)/(2\sqrt{7}) = 0.756$$

$$\cos(d'_2, d_3) = (1 + 2)/(2\sqrt{7}) = 0.567, d'_2 \text{ is } d_2 \text{ with } \textit{raise} \text{ for } \textit{hike}$$

- ▶ Words are orthogonal; documents are not. A row is a sum of one-hots, so in the **cosine similarity**  $\mathbf{z}_i^\top \mathbf{z}_j / (\|\mathbf{z}_i\| \|\mathbf{z}_j\|)$  every word two documents share adds to the numerator.
- ▶  $d_1, d_2$  share only *fed*: 0.25 on counts; 0 under TF-IDF, where *fed* weighs 0.
- ▶ Swap *hike* for *raise* in  $d_2$ : same meaning, but the cosine with  $d_3$  falls from 0.756 to 0.567.
- ▶ **Counts see shared strings, not shared meaning; Deck 5.2 repairs this.**

# Topic Models: Each Document Is a Mixture of Topics



$d_3$  with  $\theta_3 = (0.1, 0.9)$ , one draw:

hike   inflation   respond   inflation   fed  
 $z = 2$     $z = 2$     $z = 2$     $z = 2$     $z = 1$

- **Latent Dirichlet allocation** (Blei, Ng & Jordan 2003) models how the counts were written. Topic  $k$  is a distribution over words,  $\phi_k \sim \text{Dir}_V(b)$ .
- Document  $i$  has **topic shares**  $\theta_i \sim \text{Dir}_K(a)$ , summing to 1; a small  $a$  means few topics per document.
- Each token: a topic  $z \sim \text{Cat}(\theta_i)$ , then a word  $w \sim \text{Cat}(\phi_z)$ . Only the words are seen.
- **A document mixes topics; word order plays no part.**

# Fitting LDA: Collapsed Gibbs Sampling, Step by Step

toy state.  $n_k^{(v)}$ : tokens of word  $v$  in topic  $k$

|         | fed | cut | hike | support | curb | growth | inflation | respond |                     |
|---------|-----|-----|------|---------|------|--------|-----------|---------|---------------------|
| $k = 1$ | 1   | 1   | 0    | 1       | 0    | 1      | 0         | 0       | $n_1^{(\cdot)} = 4$ |
| $k = 2$ | 2   | 0   | 2    | 0       | 1    | 0      | 3         | 1       | $n_2^{(\cdot)} = 9$ |

$n_i^{(k)}$ : tokens of document  $i$  in topic  $k$

|       |   |   |       |   |   |       |   |   |
|-------|---|---|-------|---|---|-------|---|---|
| $d_1$ | 4 | 0 | $d_2$ | 0 | 4 | $d_3$ | 0 | 5 |
|-------|---|---|-------|---|---|-------|---|---|

Why it works: draws of each topic given all the others form a Markov chain that settles on the posterior. “Collapsed”:  $\phi, \theta$  are integrated out; only the  $z$ ’s are drawn.

1. Give every token a random topic  $z$ ; fill the tables.
2. Take one token (word  $v$ , document  $i$ ) out of the counts. For each  $k$ :

$$p_k \propto \frac{n_k^{(v)} + b}{n_k^{(\cdot)} + Vb} \times (n_i^{(k)} + a)$$

( $a, b > 0$  keep every  $p_k$  positive).

3. Draw its new topic from  $p$ ; put it back.
4. Every token once is a **sweep**; run hundreds. Read  $\hat{\phi}$  and  $\hat{\theta}$  off the final counts.

## Fitting LDA, Worked: One Token, Then Three Seeds

Step 2 on *fed* in  $d_3$  ( $a = 0.1, b = 0.01, V = 8$ ).  
Taking it out:  $n_2^{(\text{fed})} 2 \rightarrow 1, n_2^{(\cdot)} 9 \rightarrow 8, n_3^{(2)} 5 \rightarrow 4$ .

|         | likes <i>fed</i>    | used by $d_3$ | $p_k$       |
|---------|---------------------|---------------|-------------|
| topic 1 | $1.01/4.08 = 0.248$ | $0 + 0.1$     | <b>0.05</b> |
| topic 2 | $1.01/8.08 = 0.125$ | $4 + 0.1$     | <b>0.95</b> |

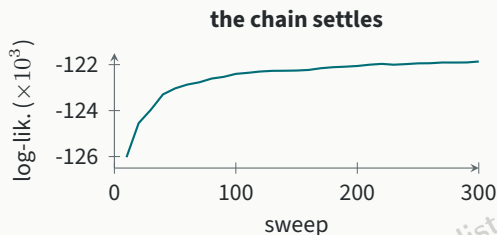
Topic 1 likes *fed* more, but  $d_3$  uses only topic 2, so *fed* almost surely goes back there.

after 200 sweeps, three seeds;  $\hat{\theta}_{i1}$  is document  $i$ 's share of topic 1

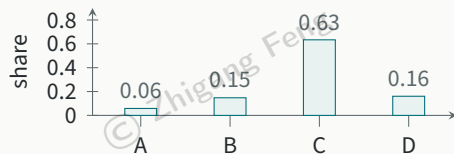
| seed | topic 1              | topic 2              | $\hat{\theta}_{11}$ | $\hat{\theta}_{21}$ | $\hat{\theta}_{31}$ |
|------|----------------------|----------------------|---------------------|---------------------|---------------------|
| 42   | fed, inflation, hike | cut, growth, curb    | 0.50                | 0.98                | 0.98                |
| 2    | cut, growth, support | fed, inflation, hike | 0.74                | 0.02                | 0.02                |
| 3    | inflation, fed, hike | cut, fed, growth     | 0.02                | 0.98                | 0.98                |

- ▶ Every seed puts  $d_2$  and  $d_3$ , the hikes, in one topic and the cut words in the other: unprompted, “tightening” and “easing”.  $d_1$ 's mix changes with the seed, and topic numbers swap (seed 3).
- ▶ **Three documents are too few for a stable fit; the FOMC run has 21,673 tokens.**

## Lab A5: Topics in the FOMC Statements, the Setup

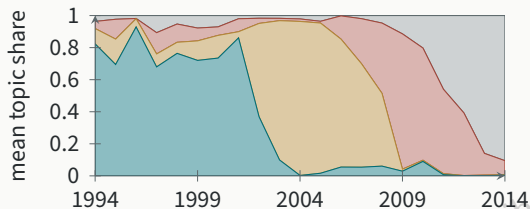


one row of  $\hat{\theta}$ : the statement of 2008-12-16



- ▶ The question: what did the 1994–2014 statements talk about? Can 4 shares replace 719 counts?
- ▶ The data: 149 statements, lower-cased; no punctuation, numbers, stop words or words under 3 letters; words in  $\geq 5$  statements and  $\leq 80\%$ : 719 words, 21,673 tokens.
- ▶ The fit: the sampler above,  $K = 4$ ,  $a = 0.1$ ,  $b = 0.01$ , 300 sweeps, about a minute.
- ▶ **The output:  $\hat{\phi}$ , four word lists, and  $\hat{\theta}$ , a  $149 \times 4$  table of shares.**

# LDA on the FOMC Statements: What It Found

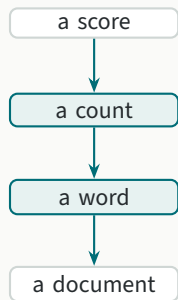


## top words, and the name we gave

- reserve, discount, board, available  
*A: discount-rate announcements*
- chairman, risks, decided, voting  
*B: the vote and the balance of risks*
- financial, markets, levels, housing  
*C: crisis, markets and housing*
- longer, term, consistent, pace  
*D: longer-run goals and purchases*

- ▶ Drafting eras as much as policy themes: A before 2002, when statements reported discount-rate approvals; B from the vote roster of March 2002; C 2008–10; D 2011–14.
- ▶ In A5: name the topics before seeing the plot, compare with a neighbour, refit with seed 7 or  $K = 6$ .  $\hat{\theta}_i$  is estimated, so its error enters any regression.
- ▶ **The model finds words that occur together; we chose  $K$  and the names.**

## When Counting Still Wins



every coordinate is a word, so a score can be traced back to its source

- ▶ When the question is how often defined words occur, transparency is the value: the uncertainty index, finance sentiment lists, political risk.
- ▶ An audit traces a score back to its words and documents; a dense vector cannot.
- ▶ Cosines of binary word vectors of 10-K product descriptions define product-market rivals (Hoberg & Phillips 2016); the same can be tried on the business sections of other annual reports.
- ▶ **Under about a thousand documents, TF-IDF is the sound baseline.**

## No Peeking: Build the Weights From Past Statements Only

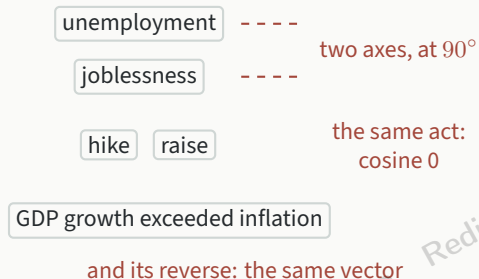
1994–2014: 149 statements      2015–2026

a 2010 statement      2020

|            | fit on 1994–2014 |           | fit on all 245 |      |
|------------|------------------|-----------|----------------|------|
| word       | df               | IDF       | df             | IDF  |
| balance    | 33               | 1.51      | 89             | 1.01 |
| securities | 49               | 1.11      | 120            | 0.71 |
| pandemic   | 0                | no column | 19             | 2.56 |

- ▶ IDF counts documents,  $\log(D/df_v)$ , and which documents you count is a choice.
- ▶ Count all 245: *balance* is in 56 of the 96 later statements, so it weighs 1.01, not 1.51, in a 2010 row. That row now depends on 2020: **look-ahead**.
- ▶ The rule: fit the vocabulary and IDF on 1994–2014 and apply them unchanged. *pandemic* gets no column; Deck 5.6 measures the cost.
- ▶ **A weight fitted on the future cannot be reproduced in real time.**

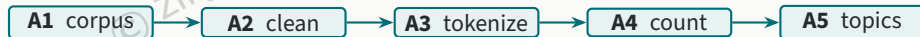
# What Counts Cannot See



- ▶ Cleaning, tokenizing and counting turned each document into a row of counts: transparent, auditable, and blind to meaning.
- ▶ Synonyms sit on perpendicular axes, word order is gone, and every word is its own axis.
- ▶ Deck 5.2 gives up one axis per word: words become points, and nearness means similarity.
- ▶ **Counts tell you which words were used, not what was meant.**

## Lab, Part A: Clean, Tokenize and Count the FOMC Statements

| cell | what you run                                          | the question you answer                                                                    |
|------|-------------------------------------------------------|--------------------------------------------------------------------------------------------|
| A1   | load 245 statements; set the 1994–2014 window         | How many statements may a fitted object see?                                               |
| A2   | five cleaning switches, one at a time                 | Which switch moves the vocabulary most, and what does it delete?                           |
| A3   | byte-pair encoding, 200 merges                        | How is a word the window never saw, such as <i>reflationary</i> , cut?                     |
| A4   | document–term matrix and TF-IDF, fitted on the window | How much held-back text falls outside the vocabulary? What would a full-sample IDF change? |
| A5   | LDA with $K = 4$ , collapsed Gibbs                    | Your names for the topics, a neighbour's, and what changes with seed 7 or $K = 6$ ?        |



► Every number on the counting slides is one you can reproduce.